

# THE SIGNIFICANCE OF DAMAGE AND DEFECTS AND THEIR DETECTION IN COMPOSITE MATERIALS: A REVIEW

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In this paper the various failure modes which occur in long fibre composites are described and discussed. The significance of each of these fracture mechanisms, in terms of their energy-dissipating capacity as well as their effect on the residual load-bearing properties, is considered. A brief review of both the destructive and non-destructive techniques used for detecting and characterizing defects and damage is presented. The ability of each technique to identify the various fracture mechanisms involved in the failure of long fibre reinforced composites is discussed and their overall suitability for damage detection evaluated.

## 1 INTRODUCTION

Fibre reinforced composite materials offer enormous potential for use in a wide number of engineering applications, ranging from sports goods to advanced aircraft structures. The superior stiffness and strength properties of long fibre composites can be utilized to manufacture complex components with lower weight at reduced cost. Composites are complex materials exhibiting distinct anisotropic properties. Fundamentally, a composite can be considered as being composed of three constituents: the fibre, the matrix, and an interphase region of finite thickness responsible for assuring adhesion between the fibre and matrix. During the manufacture of composite components, thin plies or layers of pre-impregnated fibres, typically 0.125 mm in thickness, are stacked in the desired order and the whole laminate is then processed to yield a structurally sound component exhibiting the desired physical properties. Unfortunately, the manufacturing process may result in the presence or introduction of unwanted artifacts and defects such as voids, resin-rich areas, and inclusions. Although many of these so called defects may be difficult to detect, their effects on the overall structural integrity of the component may be serious, if not disastrous.

Damage and general material degradation can also occur during the in-service operation of composite components. Typical causes of such damage are continuous cyclic loading, rapid changes in local temperature, and impact loading such as that resulting from a dropped tool or runway debris. The nature and extent of such damage will depend upon a large number of parameters including the precise loading conditions (1)<sup>†</sup>, the fibre stacking sequence (2), the properties of the constituent parts (1), as well as the prevailing environmental conditions (3). Often, damage develops over a period of months or years, and is not immediately visible to even

the trained eye. However, once the size of the defect or stress-raiser reaches a critical value, failure can be catastrophic and the consequences severe. Clearly, there is a strong need to identify as well as characterize the various types of damage and defects that occur in composite materials during manufacture and operational service. Unfortunately, there is no coherent overall design philosophy for accommodating such defects and damage in composite parts. Apart from the use of the design allowable strain limit, the approach has been generally of an ad-hoc nature.

In this paper, the various types of damage and failure mechanisms that occur in long fibre composites during both the manufacturing process as well as operational service will be examined and discussed. Following this, many of the techniques developed in order to detect failure in composite materials will be presented and their advantages and limitations discussed.

### 1.1 Manufacturing-induced defects in composite materials

During the processing stage, thin plies of either unidirectional fibres or woven cloth are stacked in a predetermined order and joined together by a chemical or fusion process to form a solid laminate. The manufacturing operation is generally a delicate procedure and great care needs to be taken to ensure that recommended manufacturing procedures are followed. However, even with the greatest care and attention, unwanted defects and stress-raisers are sometimes introduced during the critical processing stage. A detailed description of the various types of defects that may occur during the manufacture of long fibre composites has been given by Bishop (4). Some of the more common defects are listed below:

- (a) resin-rich areas either within an individual ply or at a ply interface;
- (b) voids;
- (c) distorted fibres such as ply waviness;

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<sup>†</sup> References are given in the Appendix

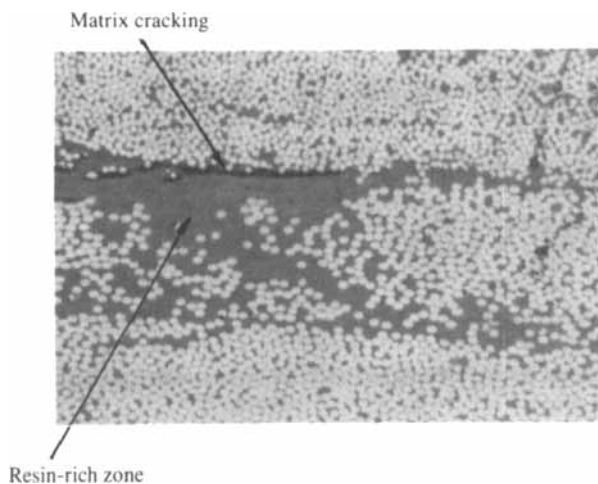


Fig. 1. Damage initiation at a resin-rich zone in an AS4/3501-6 carbon fibre epoxy subjected to an impact energy of 2 Joules

- (d) broken fibres;
- (e) inclusions such as dust or pre-preg backing paper.

Resin-rich areas are often observed in composite materials even though the part has been processed according to the manufacturer's specifications. A typical example of a resin-rich zone in an AS4 carbon fibre 3501-6 epoxy laminate is shown in the micrograph presented in Fig. 1. This section was removed from a specimen subjected to a 2 Joule impact at a location some way from the area shown in the figure. An examination of the micrograph reveals cracking in the upper part of the resin-rich zone adjacent to the fibres. This localized failure represented the first form of damage detected in this specimen suggesting that such resin-rich imperfections may be responsible for reducing the mechanical performance of composite materials. Similar conclusions have been drawn following fatigue tests on laminates containing woven fabric in which the nature of the fabric results in a resin-rich area at the cross-over points (5). When subjected to long-term fatigue loading, cracks were found to initiate within these resin-rich areas, and lead to the formation of localized planes of delamination (5).

Voiding is a problem that is common to many composite parts. The level of porosity in a composite depends upon a number of parameters including the water content, applied pressure during the cure cycle (6), and the dwell time during cure (7). Also, voiding is usually more common at angles or corners of composite parts. Tomasino (8) showed that the out-of-plane tensile strength of a quasi-isotropic laminate decreased rapidly with increasing porosity, Fig. 2. Hancox (9) studied the effect of low (<1 percent) and high (>5 percent) void contents on the torsional properties of CFRP exposed to both dry and wet environments. He showed that specimens with high voiding absorbed considerably more water than those with low void contents. Mechanical testing revealed that excessive voiding reduced the interlaminar shear strength of the composite by approximately 30 percent.

The processing step, particularly in thermoplastic-based composite materials such as carbon fibre reinforced PEEK (APC2), can often result in the formation

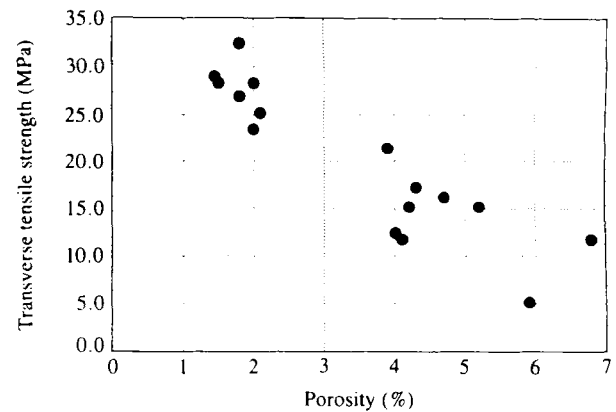


Fig. 2. The variation of transverse tensile strength with porosity in an AS4/3501-6 carbon fibre epoxy composite (reference 8)

of fibre kinks or ply waviness (10). Such distortion of the principal load-bearing constituent may occur for a number of reasons. Firstly, the coefficient of thermal expansion of the carbon fibres and polymer matrix are very different, leading to the build-up of residual stresses on cooling. Also, the process of crystallization in semi-crystalline PEEK results in a volume contraction of approximately 18 percent (11). This reduction in the matrix volume may generate a compressive stress field around the fibre and lead to fibre kinking. It is also likely that rapid cooling of the composite induces residual stresses due to the reduced time for material relaxation (12). Raman spectroscopy studies on HMS4 carbon fibre/PEEK have shown that the strains developed in the fibres during the processing cycle can be as high as 0.28 percent (13). The effect of fibre waviness on the mechanical properties of CFRP has been investigated by van Dreumel and Kamp (14). Their results show that fibre kinking does not affect the tensile modulus of unidirectional CFRP, but it can result in a 20 percent loss in tensile strength.

Residual stresses resulting from the cure or processing cycle can also be responsible for localized matrix cracking in multi-directional laminates, either in the matrix or the fibre-matrix interphase region (12). Such cracks are frequently large, sometimes traversing a whole ply, as shown in the optical micrograph presented in Fig. 3. The effect of such damage on the mechanical properties of the laminate can be significant, and will be discussed in more detail in the following section.

As stated previously, the fibres in a composite material are responsible for carrying the majority of the applied load. Consequently, if these fibres are damaged or fractured, either in the handling or processing stage, the strength of the component may be greatly reduced. Work by Rhodes (15) on a (0 degrees  $\pm$  45 degrees) CFRP composite showed that a single cut tow caused a 25 percent reduction in tensile strength. In compression, the loss in strength was less (approximately 11 percent), suggesting that this form of loading is less sensitive to the presence of damaged fibres.

The inadvertent inclusion of a piece of pre-preg backing paper is not an unknown occurrence. The presence of such an inclusion may well have serious consequences on the subsequent mechanical performance of

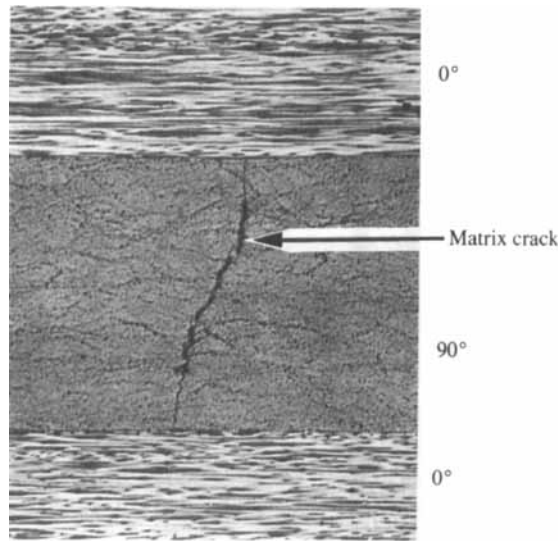


Fig. 3. A matrix crack in a carbon fibre epoxy composite resulting from excessive thermal stresses

the laminate. Rhodes (15) introduced a 10 mm diameter paper inclusion under a 0 degrees ply in a (0 degrees  $\pm$  45 degrees) carbon fibre honeycomb sandwich structure and found that the compressive strength of the part was reduced by up to 25 percent with failure occurring in a catastrophic gross delamination-type mode.

## 1.2 Summary

Clearly, care has to be exercised when manufacturing composite components in order to avoid unwanted stress-raisers and defects. A summary of the effects of the various manufacturing-associated defects is given in Table 1. Voiding gives, perhaps, the greatest cause for concern since it is common to many components and has a serious effect on certain fundamental mechanical properties. Such porosity also results in a greater level of water absorption which may further reduce the integrity of the laminate.

## 2 FAILURE MECHANISMS IN COMPOSITE MATERIALS

Previous studies (1)–(3) (5) (16)–(18) have identified a large number of failure mechanisms in composites. These include: intralaminar matrix cracking, plastic flow, delamination fibre–matrix debonding, fibre pull-out, and fibre fracture. The relative contribution of each during fracture will depend upon a large number of parameters.

Table 1. The effect of manufacturing-induced defects on the mechanical properties of composites

|                 | Mechanical property  | Effect             |
|-----------------|----------------------|--------------------|
| Fibre waviness  | Tensile modulus      | No loss (14)       |
|                 | Tensile strength     | 20% loss (14)      |
|                 | Poisson's ratio      | 100% increase (14) |
| 10% voiding     | Compressive strength | 15% loss (6)       |
|                 | ILSS                 | 30% loss (6)       |
|                 | IL shear modulus     | 30% loss (6)       |
| Paper inclusion | ILSS                 | 25% loss (15)      |
|                 | Compressive strength | 20% loss (15)      |
| Cut fibre tow   | Tensile strength     | 25% loss (15)      |
|                 | Compressive strength | 11% loss (15)      |

In the following sections, the significance of these fracture processes will be considered, and their effects on the subsequent mechanical performance evaluated.

The fracture process of a composite material involves both macroscopic as well as microscopic failure mechanisms. Both are extremely important in terms of energy dissipation in composite materials. A schematic representation of local failure processes as a crack propagates in a long fibre composite is presented in Fig. 4. Here, damage may involve failure of the fibre–matrix bond, fibre fracture, and plastic deformation, and failure of the matrix.

### 2.1 Fibre–matrix debonding

When the stress in the fibre–matrix interphase exceeds the local strength, debonding occurs and a crack forms. Debonding represents, therefore, a very localized mode of failure that is often very difficult to detect using conventional techniques. The amount of debonding present within a composite depends upon the level of surface treatment applied to the fibres during the manufacture of the pre-preg. Generally, fibres with low levels of surface treatment tend to debond more easily and the resulting fracture surfaces are usually rough and strongly three-dimensional when viewed in a scanning electron microscope (16). Highly treated fibres debond less and fracture tends to be planar with cracks propagating directly across fibres (16).

Beaumont (17) gives the energy for debonding in a long fibre reinforced composite as

$$W_d = \frac{\pi d^2 \sigma_f^2 l_d}{24 E_f}$$

where

- $d$  = the fibre diameter
- $\sigma_f$  = the tensile strength of the fibre
- $l_d$  = the length of the debond
- $E_f$  = the modulus of the fibre

Increasing the level of surface treatment applied to the fibres reduces the debond length,  $l_d$ , reducing the fracture energy of debonding. Kirk *et al.* (18) determined a value of 6 KJ/m<sup>2</sup> for the debonding energy of a carbon fibre/epoxy.

Once the fibre has debonded, further loading results in differential displacement between the fibre and matrix, and a frictional force at the boundary between the two constituents. The work done per fibre in post-debonded friction can be written as

$$W_{pdf} = \frac{\pi d \tau l_d^2 \epsilon_f}{2}$$

where

- $\tau$  = the frictional shear stress
- $\epsilon_f$  = the fibre strain to failure

In fibreglass reinforced composites, post-debond sliding represents one of the principal energy absorbing mechanisms, and is in part responsible for the relatively high toughness associated with fibreglass-based composite materials (17).

Fibre–matrix debonding does not appear to have a significant effect upon the load-carrying capability of a

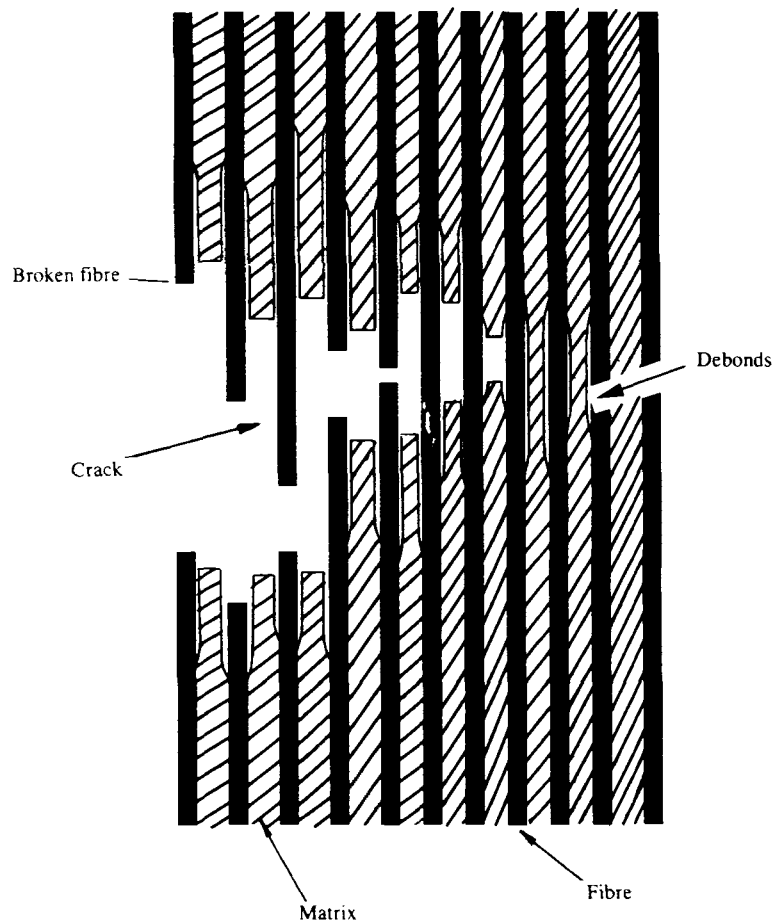


Fig. 4. Schematic representation of crack propagation in a fibre reinforced composite.

composite. Indeed, in certain cases it may have a desirable effect since debonding can reduce the stress concentrating effect of a sharp crack or similar defect. Debonding does, however, permit the ingress of water into a composite which may, in turn, have a negative effect particularly on the compressive properties of a composite component.

## 2.2 Fibre pull-out

In a region of high stress concentration, such as at the tip of an advancing crack, fibres often fail and fracture. As the crack front continues to advance, these fibres are pulled out of the surrounding matrix. In overcoming the resultant frictional force, work is done and energy is dissipated. If one assumes that the maximum fibre pull-out length is equivalent to one-half of the critical fibre length  $l_c$  then the work to pull-out a fibre is given as (17)

$$W_p = \frac{\pi d \tau l_c^2}{24}$$

Surface treating the fibres reduces the critical transfer length resulting in a reduction in the energy dissipated in fibre pull-out (17). Nevertheless, fibre pull-out is one of the primary energy dissipation mechanisms in carbon fibre reinforced epoxies. Kirk *et al.* (18) reported a value of 110 KJ/m<sup>2</sup> for the pull-out energy of a carbon fibre/epoxy, and Bandyopadhyay *et al.* (19) a rather higher

value of 800 KJ/m<sup>2</sup> for a carbon fibre/bismaleimide composite.

## 2.3 Intralaminar matrix cracking

Intralaminar matrix cracking is a very common mode of failure in polymer matrix composites. In this work, matrix cracking includes splitting, a term referring to long cracks parallel to the fibres, either in the matrix or within the interphase region. An example of a large matrix crack resulting from an excessive residual stress field is shown in Fig. 3. Localized matrix cracking in a carbon fibre epoxy composite end notch flexure (ENF) specimen following mode II loading is shown in Fig. 5 in which matrix cracking in front of the primary mode II crack is evident. These localized matrix cracks may eventually coalesce to form a primary crack. The fracture energy associated with matrix cracking is likely to be low, typically several hundreds of J/m<sup>2</sup> for a brittle carbon fibre/epoxy (20)–(21), and several thousand J/m<sup>2</sup> for a thermoplastic-based composite (22). Although matrix cracking is frequently very localized, and often extremely difficult to detect, it can, under conditions such as fatigue loading, act as a precursor to delamination – a more detrimental form of damage. During impact on a flexible target, matrix cracking initiates at the lower surface of the target and propagates upwards through the laminate forming planes of delamination (23). Matrix

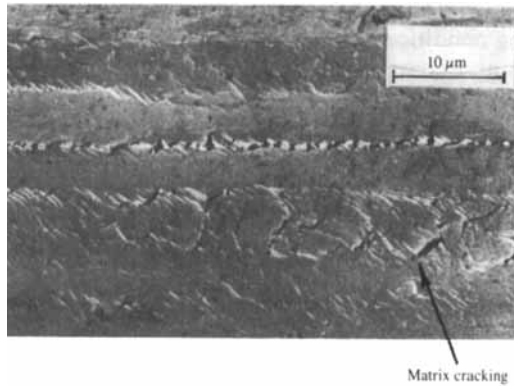


Fig. 5. Localized matrix cracking at the tip of a mode II crack in AS4 carbon fibre PEEK (reference 34)

cracking frequently deflects planes of delamination from one ply interface to another, and is in part responsible for the large areas of fracture observed following transverse impact loading. A typical example is shown in Fig. 6 in which the optical micrograph of a carbon fibre/epoxy composite subjected to an impact energy of 2.4 Joules is presented. Planes of delamination spreading well away from the impact are evident. In all cases, the delaminations are linked by a network of matrix cracks inclined at approximately 45 degrees.

The effect of matrix cracking on the mechanical properties of fibre-reinforced composites has received very little attention. Bishop and Dorey (24) showed that matrix cracking between the fibres in a 45 degrees ply may impose a significant stress concentration on neighbouring 0 degrees fibres which, in turn, may result in premature failure of the complete laminate. A detailed analysis by Kriz (25) showed clearly that matrix cracks in 90 degrees plies of a moisture-saturated (0 degrees, 90

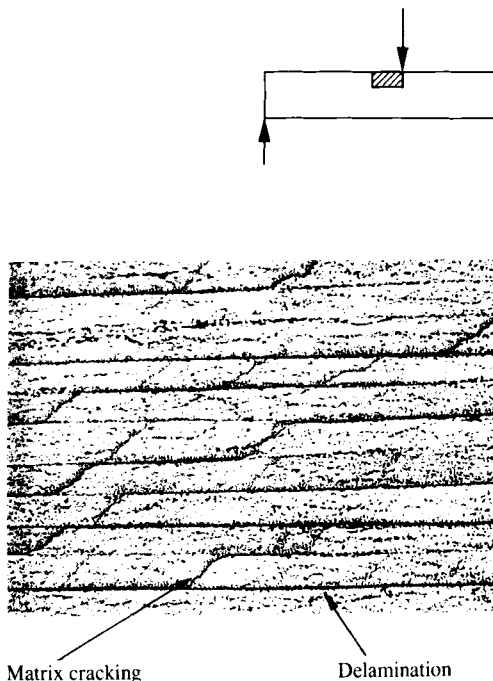


Fig. 6. Planes of delamination linked by matrix cracking in an impact damaged XAS/914C CFRP laminate

degrees  $\pm 45$  degrees) carbon fibre/epoxy composite precipitated failure in adjacent 0 degrees plies. It was also shown that the failure locus of the 0 degrees plies could be related directly to the position of matrix cracks in the 90 degrees plies. This is illustrated in the optical micrograph presented in Fig. 7 in which a matrix crack in the 90 degrees ply resulted in a number of fibre fractures in the adjacent 0 degrees ply. The occurrence of such cracks was subsequently shown to reduce the tensile strength of the laminate by almost 10 percent.

Matrix cracking can have a positive effect, under certain circumstances. Kellas *et al.* (26) showed that matrix cracks between the 0 degree plies in a (0 degrees  $\pm 45$  degrees) CFRP laminate reduced the stress concentrating effect of a sharp notch, resulting in an improved tensile strength.

## 2.4 Matrix deformation

Plastic deformation in the matrix of a polymer-based composite is rarely discussed when considering failure processes in advanced composites. It is, however, a very important deformation process, and is largely responsible for the high toughness characteristics exhibited by many of the current generation of thermoplastic composites. Simple tension tests on pure PEEK samples have shown that the strain to failure may be as high as 100 percent; more than an order of magnitude

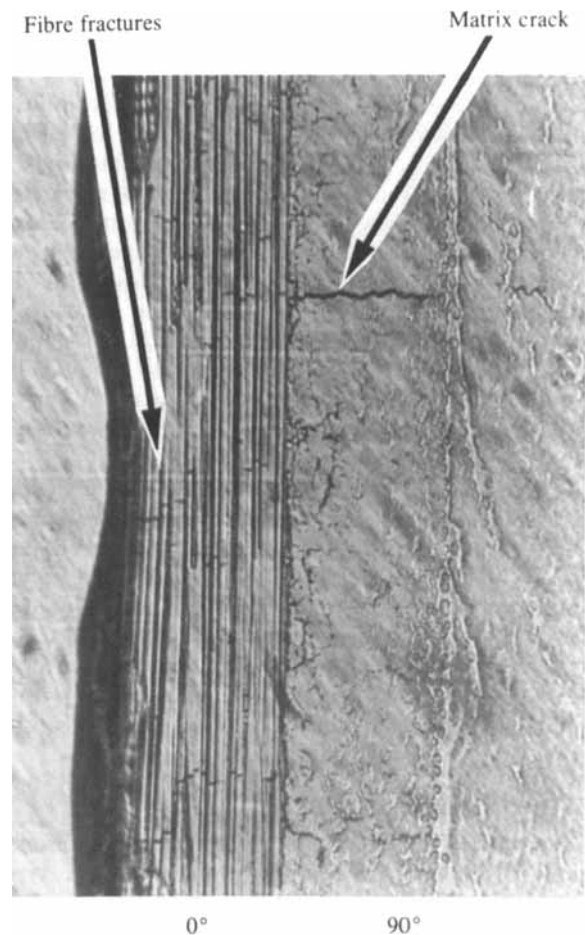


Fig. 7. Localized fibre fracture resulting from the presence of matrix cracking in a carbon fibre epoxy laminate (Kriz)

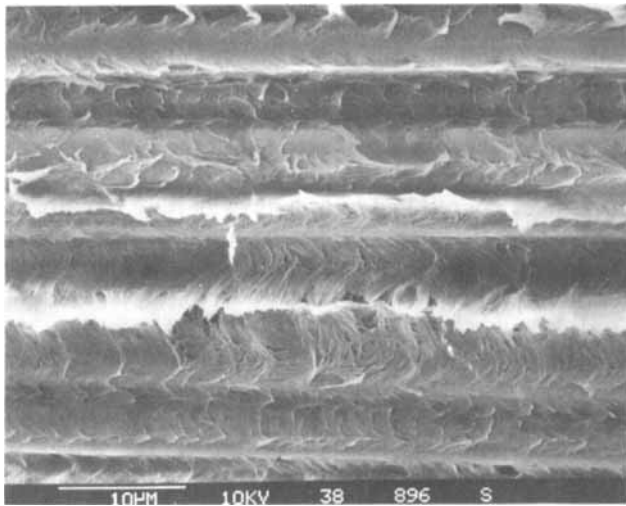


Fig. 8. Scanning electron micrograph of the mode I fracture surface of an IM6 carbon fibre PEEK DCB specimen

greater than many of the epoxies used to fabricate composite laminates. Hirschbuehler (27) showed that matrices with high strains to failure offered excellent compression after impact properties, when used in long fibre reinforced composites. The amount of shear deformation that has occurred during failure can be determined by examining the fracture surface in a scanning electron microscope (SEM). The fracture surface of a carbon fibre reinforced PEEK specimen tested under mode I loading conditions is shown in Fig. 8. Examination of the fracture surface indicates that the matrix has been drawn considerably, a process involving considerable dissipation of energy. Shear flow in composites is likely to be highly desirable since it has the effect of blunting sharp cracks, resulting in a redistribution of the local stress field. Double cantilever beam tests on carbon fibre/PEEK have shown that the fracture toughness corresponding to crack initiation is higher in specimens with a greater ability to undergo plastic flow (28). Few workers have attempted to quantify or measure the energy associated with matrix deformation in composite materials. Hine *et al.* (29) reported a value of 1.39 KJ/m<sup>2</sup> for carbon fibre/PEEK at room temperature. Values for more brittle epoxy-based composites are likely to be an order of magnitude below this.

At high rates of strain, such as that imposed by localized impact loading, the yield stress of the polymer is usually significantly higher than that measured under quasi-static loading conditions. The ability of a polymer to undergo plastic deformation is, therefore, reduced and its toughness diminished. Dan-Jumbo *et al.* (30) showed that carbon fibre/PEEK, a tough matrix composite, underwent a transition at high rates of strain, with the material exhibiting a relatively poor energy-absorbing capacity under these conditions.

## 2.5 Delamination

Delamination is one of the most frequently discussed modes of failure in composite materials. A typical example of delamination fracture is shown in the impact-damaged CFRP specimen presented in Fig. 6. Small areas of delamination are capable for reducing the compression strength of composite materials by over 50

percent (4). Delamination occurs under a wide range of loading conditions such as in-plane quasi-static loading, tensile and compressive fatigue loading (1), and impact loading (31).

Delaminations propagate at ply interfaces in multi-directional long fibre composites. Liu (32) has shown that under low velocity impact loading conditions delamination is most severe at interfaces at which the difference in relative angle between the upper and lower plies is greatest.

Typically, the fracture energy of delamination varies between 100 J/m<sup>2</sup> (epoxy-based composites) (33) to 3000 J/m<sup>2</sup> (thermoplastic-based composites) (34). Since the fracture energy for this mode of failure is strongly dependent upon the ability of the matrix material to undergo shear flow, it is again rate dependent and may drop by up to an order of magnitude at high rates of strain (35).

As stated previously, delamination has serious consequences on the compressive properties of long fibre composites. This form of *interlaminar* fracture reduces the stability of the load-bearing fibres resulting in a localized buckling-type of failure mode at low loads.

In recent years, a concerted effort has been made by the manufacturers to improve the toughness characteristics of matrix materials. At present, toughening of traditionally brittle epoxy resins is achieved in a number of ways; the addition of an elastomeric phase such as CTBN particles, the addition of a thermoplastic phase such as polysulphone or polyethersulphone, or by varying the crosslink density. Many composite producers are now using tough thermoplastic matrices such as PEEK and PES in order to achieve greater resistance to delamination and, therefore, improved post-impact compressive properties.

Delamination is not, however, always detrimental to the load-carrying capability of a composite structure, indeed, in certain cases it may be beneficial. One example of this has been identified by Bishop (4), following fracture tests on multi-directional CFRP laminates. Here, it was noted that the presence of zones of delamination around a stress-raiser such as a sharp notch served to re-distribute the stress field as well as isolate the defect.

## 2.6 Fibre fracture

Since the fibres represent the principal load-bearing constituent of a fibre reinforced composite, fibre fracture can have a severe effect upon both the stiffness and strength of a multi-directional composite (31). Fibre damage in composite materials may occur for a number of reasons. Transverse impact loading often creates zones of localized fibre fracture immediate to the point of impact. Compression fatigue cycling also has been shown to result in the generation of large angled cracks in the 0 degree fibres in a (0 degrees  $\pm$  45 degrees) laminate, Fig. 9 (1). It has been suggested that the fracture of even a very small number of fibres in a composite component may be sufficient to precipitate failure (4). Fibre fractures are likely to be most detrimental to the tensile strength of a composite coupon or part.

Several workers have attempted to measure the energy required to fracture carbon fibre reinforced composites in a transverse mode (26) (36) (37). Dorey (36) quoted values of 20 and 60 KJ/m<sup>2</sup> for the transverse fracture energies of treated and untreated CFRP composites

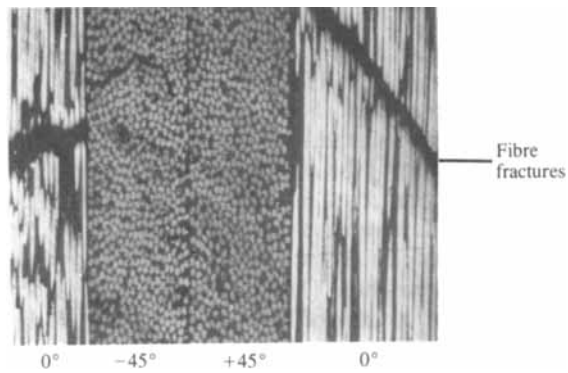


Fig. 9. Optical micrograph of an XAS/914C specimen following 0-compression fatigue loading (S. Kellas)

respectively. Leach and Moore (22) measured transverse fracture energies as high as  $406 \text{ KJ/m}^2$  for carbon reinforced PEEK. Clearly, the fracture energy associated with this mode of failure is considerably greater than those relating to matrix-dominated mechanisms. Energy can, therefore, be dissipated in either small areas of fibre fracture or large zones of matrix damage. Impact perforation tests on a brittle carbon fibre/epoxy have shown that transverse fibre fracture is the principal energy-absorbing mechanism in thin composites, whereas matrix-dominated models of failure become equally important in thicker laminates, Fig. 10.

### 2.7 Summary

Amongst the failures modes presented above, delamination-type damage gives the greatest cause for concern since large areas of this form of interlaminar fracture can be generated under low energy impact conditions. When loaded in compression, the stability of the load-bearing fibres is reduced significantly and failure may result at low loads.

Other forms of damage, such as fibre fracture and matrix cracking, are also detrimental to the residual performance of these advanced materials. Although the extent and severity of damage incurred may be less severe than delamination, they can, nevertheless, reduce both the short and long-term strength of the material significantly.

There exists, therefore, a clear need to identify and characterize damage in engineering composites at the

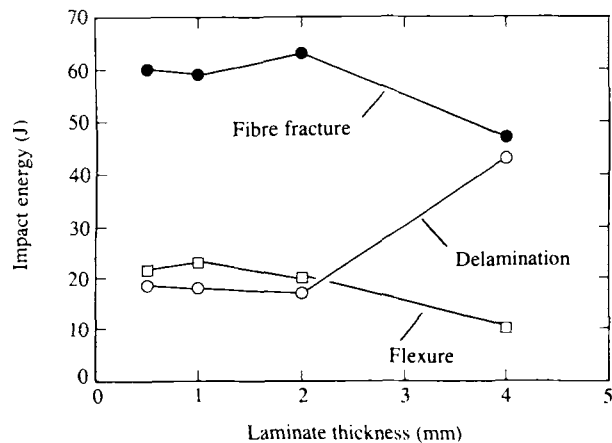


Fig. 10. The calculated dissipation of energy in a series of perforated ( $\pm 45$  degrees) CFRP specimens. Low velocity impact loading (reference 31)

earliest possible opportunity. The ideal damage detection technique should be simple to use and be capable of identifying all of the failure modes observed in composite materials. At present, a number of NDE techniques exists. Unfortunately, no technique is universal in its ability to identify all types of damage and defects. In the following section, the more commonly used damage detection techniques currently employed will be presented and their capabilities discussed. The discussion will not be limited solely to non-destructive evaluation techniques, but will also consider the range of destructive techniques used by research workers to identify and characterize damage.

## 3 NON-DESTRUCTIVE DAMAGE DETECTION TECHNIQUES

By far the simplest non-destructive damage evaluation (NDE) technique is visual examination. A simple inspection of the damage zone can often yield useful information regarding the location as well as the severity of any defects. For example, impact loading of carbon fibre PEEK leaves a dent on the impacted surface (38). This dent is usually readily identifiable and can be used to give an indication regarding the level of damage within the volume of the laminate.

Many of the NDE techniques currently available for inspecting composites have existed for a number of years, having been developed originally for use on metallic structures. In the following sections, the more commonly used techniques for detecting damage and defects in composites will be reviewed. The discussion is by no means exhaustive, but is designed to present the basic techniques as well as their capabilities and limitations. More detailed descriptions of the principles and theory behind the techniques are given elsewhere in the literature (39)–(42).

### 3.1 Ultrasonics

Ultrasonic inspection is one of the most commonly used techniques for detecting defects and damage in laminated composites. Here, ultrasonic signals are directed towards the component under inspection. In order to ensure maximum transmission of the ultrasonic energy, the test specimen is either immersed in water, coated with a grease or gel, or the signal is transmitted through a jet of water. The quality of a composite panel is assessed by placing a transducer at the rear of the panel (through-transmission) or by the same transducer that emitted the original pulse (pulse-echo detection). In either case, the ultrasonic signal received is converted into an electrical signal, amplified, and then displayed as a vertical signal on an oscilloscope, the horizontal axis being a time-base. Attenuation of the ultrasonic signal occurs as a result of visco-elastic effects in the composite matrix, geometric dispersion due to the heterogeneity of the laminate as well as dispersion due to damage or defects within the material (40). By proper selection of the signal frequency, visco-elastic and heterogeneity effects can be minimized and attenuation resulting from internal damage maximized.

Three modes of application can be undertaken, A, B, and C scans. The A scan gives information concerning the quality of a component at a single point. The amplitude of the arriving signal, and its position relative to



those of the signals corresponding to the upper and lower surfaces of the target, give an indication of the severity and the through-thickness location of the damage or defect. The B scan is essentially a linear collection of A scans and is, therefore, equivalent to taking a slice through the sample. The third, and perhaps the most useful, type of ultrasonic technique is the C scan method in which the component is placed above a glass plate immersed in a bath of water, and the transducer sweeps back and forth across the component, receiving and analysing the signal reflected from the upper surface of the glass plate. The data are analysed by time gate which converts the amplitude of the largest signal into a voltage, and the information presented on a current sensitive recording paper. The C scan data are generally presented as a function of shade or colour. In one mode of data presentation, sound areas of the composite laminate are presented as black, severely damaged as white, and intermediate levels as varying shades of grey.

The C scan technique suffers from several limitations, such as the fact that the tank may have to be large in order to accommodate certain structural components. Further, the water/composite impedance ratio is much smaller than that of air/composite, so care has to be taken in order to ensure that water does not enter damage zones (40). Honeycomb structures float and are difficult to immerse. These problems may be overcome, to a large degree, by using a water jet system, where the signal is transmitted to the part in a jet of water ejected from a nozzle located just above the surface of the laminate.

The ultrasonic C scan technique is ideally suited to the detection of delamination-type fracture such as that introduced as a result of impact loading (41). The test is quite sensitive, being capable of determining the size of a

defect in a 2 mm thick composite to within  $\pm 0.64$  mm (43). The C scan analysis cannot detect fibre fracture, matrix cracking or the micro-mechanical damage mechanisms reported above. Further, the C scan technique, when in its standard mode, yields only a two-dimensional view of the defect or damage zone, giving no through-thickness data, as shown in Fig. 11(a) in which the damage developed around a circular notch in a carbon fibre/epoxy coupon after reversed axial fatigue loading is documented (1). Greater information can be obtained from a time-of-flight analysis in which data from the A and C scan techniques are combined to yield a three-dimensional representation of the damage zone. An example of the time-of-flight analysis is shown in Fig. 11(b). (44).

Other recent advances in ultrasonics technology have led to the development of systems with multi-element transducers, and roller probes suitable for rapid detection of impact damage under dry conditions (41).

### 3.2 X-ray radiography

Radiography depends upon the differential absorption of radiation by the specimen or component under examination. The level of unabsorbed radiation that has passed through the coupon is monitored on a fluorescent screen, film, or photo-sensitive paper. In order to improve the level of contrast between defects and the base material, a penetrant is usually applied to the coupon before examination. The choice of penetrant depends upon the level and type of damage within the composite. However, zinc iodide is frequently used since it is readily available, does not influence the mechanical properties of the laminate, and is not toxic.

A penetrant-enhanced X-ray radiograph of an impact-

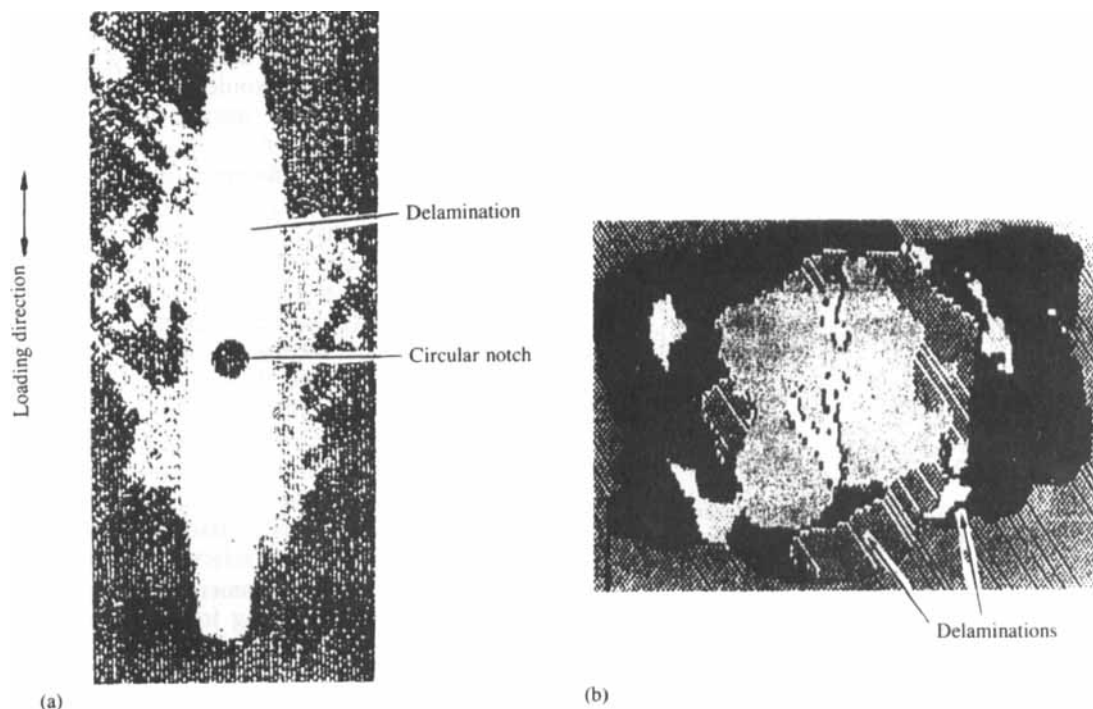


Fig. 11. (a) Typical C scan record of a fatigue-loaded notched CFRP specimen (reference 1). (b) Time-of-flight C scan of an impact-damaged CFRP laminate. Each shade of colour corresponds to damage at a different level (reference 44)



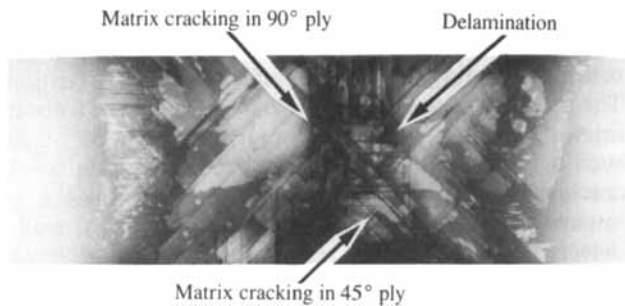


Fig. 12. Penetrant-enhanced X-ray radiograph of an impact-damaged CFRP specimen (Kellas)

damaged (0 degrees, 90 degrees  $\pm$  45 degrees) CFRP specimen is shown in Fig. 12. X-ray radiography is an extremely useful technique for identifying damage such as matrix cracking, delamination, and extensive fibre fracture. If a fine grain X-ray film and a relatively high voltage are used, defects considerably less than 1 mm in size can be detected (45) (46). In regions where damage is severe, the resolution of the technique is limited, and differentiating between the various failure modes becomes difficult, if not impossible. Penetrant-enhanced in-situ radiography enables real-time analysis of the processes involved in damage development and can yield considerable information regarding the mechanisms of deformation and failure in composites (47). A three-dimensional appreciation of the damage zone can be obtained by taking two radiographs of the composite, one normal to the beam and the other at an angle to the X-ray beam. When viewed through a stereo viewer a three-dimensional image is obtained.

The X-ray radiographic technique does not lend itself to general production inspection of manufactured components, but is more suited to high-resolution damage characterization.

### 3.3 Thermography

The thermographic technique depends upon the differential absorption and dissipation of heat in a damaged composite component. Two types of analysis are used, namely passive and active thermography. The former relies on an external heating source and the latter on the internal heat generation from friction and fretting of fracture surfaces. The latter technique is suited only to fatigue-loaded structures where a continuous heat generation process occurs. In the passive method, the surface of the component is subjected to a rapid temperature rise and the subsequent heat flow monitored. Since the flow of heat across a damage zone is reduced, defects such as impact damage result in the accumulation of heat, which can be detected by an infra-red sensitive camera.

The effectiveness of the technique depends upon the thermal conductivity of the material under examination. In the case of carbon fibre composites, the thermal conductivity in the plane of the laminate is approximately nine times higher than that in the through-thickness direction (41). As a result of this low transverse thermal diffusivity, defects or damage zones located at or near the centre of a thick composite are difficult to detect.

Thermography is best suited to the detection of gross delaminations such as those introduced by localized impact loading. The degree of resolution depends, to a large extent, upon the thermal properties of the component as well as upon its thickness. However, thermography is less sensitive than the various ultrasonic methods. One of the major advantages of thermography is that it can be used for examining large structures such as composite wings or fuselage sections.

A variation on this technique is vibrothermography in which low amplitude vibrations are applied in order to induce localized heating in a structure, and the data collected via an infra-red camera (48). An example of such a thermograph is shown in Fig. 13, in which an image of a subsurface circular defect in a multi-directional CFRP laminate is shown (49).

### 3.4 Acoustic emission

Acoustic emission relates to the generation, propagation and detection of stress waves in materials as they undergo deformation and fracture. These waves propagate to the surface of the material where they are detected by a transducer. In general, acoustic events are precipitated by applying a moderate stress to the component. The technique is capable of detecting most of the failure modes associated with composite materials but suffers the disadvantage that it is not easy to differentiate between them (3). Following fatigue tests on a carbon fibre epoxy, Cohen and Awerbach (50) concluded that the emissions caused by friction and rubbing of fracture surfaces within composites exceeds, and, therefore, may conceal emission activities associated with the actual fracture process. Detailed studies by a number of workers suggest that it may be possible to isolate matrix cracking (51), fibre breaks (52), and fibre-matrix debond-



Fig. 13. Image of a CFRP laminate containing a central defect as detected by vibrothermography (reference 49)

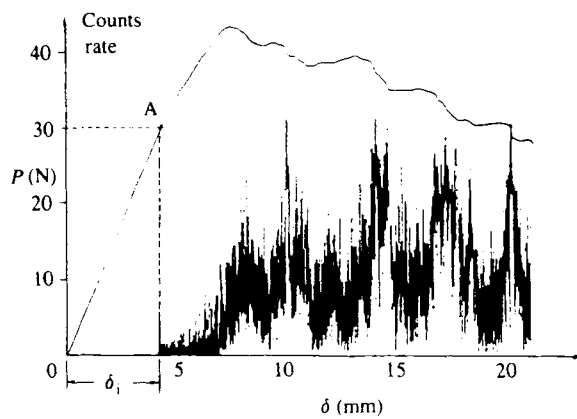


Fig. 14. Load and acoustic emission counts versus displacement for a glass-polyester composite (reference 54)

ing (53). Unfortunately, the acoustic emission technique is hindered by the fact that composite materials tend to attenuate and disperse propagating stress waves. As a result, it is usually necessary to use a number of transducers on large or complex structures. The acoustic emission technique is perhaps best suited to the proof testing of composite structures or detecting damage initiation in composite parts. An example of the latter is shown in Fig. 14 in which the acoustic emission spectra corresponding to a mode I test on a double cantilever beam are presented (54). Here, the onset of acoustic activity is shown to coincide with a change in slope of the load-displacement curve, a point generally taken as corresponding to damage initiation in such specimens.

One recent development of this procedure is the thermo-acoustic technique (55) in which the composite is heated either in an oven or by a hand-held heater, and the acoustic emissions recorded using a conventional piezo-electric transducer. Sato *et al.* (55) showed that significant acoustic events were recorded only in components containing internal damage. It was suggested that emissions occur as a result of sliding of crack interfaces induced by redistribution of the residual stresses. The technique has been used to detect both impact as well as fatigue damage in composite parts.

### 3.5 Eddy currents

Eddy currents can be used to detect fibre damage in composites containing electrically conducting fibres. The technique is based on the induction of an electrical field in the composite by a current-carrying coil positioned just above the surface of the component. Flaws within the material interrupt the current field resulting in change of impedance in the coil. The severity of the defect is characterized by the change in phase angle and amplitude of the electrical signal. The eddy current technique cannot detect delamination (56) being suitable only for detecting fibre fracture in composites. It is best used in conjunction with an ultrasonic technique such as the C scan, enabling both delamination and fibre damage to be detected.

### 3.6 Fibre optics

The use of fibre optics in advanced composite structures permits continuous or semi-continuous evaluation of the integrity of the composite structure to be made. The

optical fibres are introduced into the composite during the manufacturing stage and the whole component processed according to the manufacturer's specifications. The integrity of the composite part is evaluated from the intensity of light passing through the fibres (57). When such a fibre is damaged or fractured as a result of the cracking or localized failure within the composite, the intensity of light at the output station drops, and is detected by a sensor. The technique is particularly promising since it offers a quick, reliable means of evaluating the state of a component. The greatest sensitivity is obtained when the fibres are placed perpendicular to the reinforcing fibres in the surrounding material (58). The optimum through-thickness location of the fibres depends upon the projected operational conditions. If the component is expected to be subjected to flexural loading, then the fibres should be positioned nearer the lower surface (58). Unfortunately, the optical fibres are usually considerably larger than the reinforcing fibres and may result in ply deviation (59), or act as stress-raisers precipitating delamination (41). Also, the strength of the interface between the optical fibres and the composite matrix has to be sufficiently high in order to avoid unwanted cracking (60). Care has to be exercised in order to ensure that the failure strain of the optical fibres is not greater than that of the material constituents otherwise damage may go undetected (58). The technique is capable of detecting many of the matrix as well as fibre-dominated modes of failure and has been shown capable of detecting damage consistent with barely visible impact damage (58).

### 3.7 Holography

The theory behind holographic non-destructive testing is beyond the scope of this review and discussed in detail elsewhere in the literature (61). Holography is a technique by which the image of a three-dimensional object is stored on a photographic emulsion. If the test specimen is slightly distorted and a second exposure made upon the emulsion the light states emanating from the two exposures interfere to produce interferometric fringes (62). These fringes are equivalent to lines of equal elevation with each fringe representing the locus of points displaced out-of-plane by one-half wavelength of the light used to produce the hologram. By observing abrupt changes in the fringe pattern, damage such as delamination, surface matrix cracks and surface fibre failures can be detected (40). More recent developments using a compact TV system have enabled real-time holographic measurements to be made on composite parts (63).

### 3.8 Moiré interferometry

The principles of moiré interferometry were first outlined by Guild (64), and the technique applied to the assessment of the severity of defects in composites by Marchant and Bishop (65). A full description of the theory and practical application of moiré interferometry is presented by Czarnek (66).

Moiré interferometry is an optical technique which provides contour maps or fringe patterns representing the in-plane displacement components of surface deformation. A fine crossed line diffraction grating is applied to the surface of interest, and the body loaded and

viewed in an interferometer. The deformed grating diffracts the two coherent beams which are used to illuminate the specimen. The diffracted beams are then focused on a photographic plate, where they interfere to produce a fringe pattern corresponding to one of the in-plane displacement components. The sensitivity of the method depends upon the optical arrangement used, but a typical value is  $0.417 \mu\text{m}$  per fringe. This is a powerful quantitative strain analysis tool which was originally developed for laboratory use, but which has now been refined for field application (67) (68). The disturbance in the displacement fields due to the presence of an embedded fibre optic, mentioned above, has been evaluated using moiré interferometry (69).

Moiré interferometry can also be used to detect and monitor the development of matrix cracking and delamination in composite materials. The effects of these forms of damage are discontinuities in the fringe patterns and changes in the shape and density of the fringe patterns as the damage develops, and loads are redistributed in the specimen. An example of this is shown in Fig. 15 in which matrix cracks are first apparent in the 90 degrees plies on the edge of a quasi-isotropic carbon fibre/epoxy coupon (70). As the applied load increases additional cracks are documented in the 90 degrees plies. There is also evidence of some cracking in the outer 45 degrees plies. Further loading causes extensive delamination between the central 90 degrees plies and the adjacent 0 degrees plies.

Moiré interferometry should not be confused with shadow moiré which can also be used in damage monitoring. Shadow moiré provides fringes representing the out-of-plane displacement with a sensitivity one or two orders of magnitude less than moiré interferometry. Mousley (71) used shadow moiré to monitor the growth

of delamination damage initiated by an implant in a compressively loaded panel.

### 3.9 Other NDT techniques

Other techniques used for detecting and evaluating damage in composite materials include X-ray tomography, the coin-tap test, leaky Lamb waves, D sight and edge replication. X-ray tomography is in many ways similar to the X-radiographic technique outlined above. The principal difference between the two is that X-ray tomography yields a three-dimensional view of the damage zone, whereas conventional X-radiography presents a two-dimensional image on a film or photographic paper. The technique was developed for the medical industry but has recently been used to detect porosity, air entrapment and delamination in composite materials (72).

The coin-tap test has been adapted and applied to composites by Adams *et al.* (73). A small solenoid-actuated hammer with a force gauge incorporated in its head is accelerated towards the surface of the part under examination. The force-time pulse is measured, compared to a pulse corresponding to a structurally-sound region, and the difference plotted on electro-sensitive paper in a similar fashion to ultrasonic C scanning. One of the main advantages of the coin-tap procedure is that it requires no couplant, unlike the ultrasonic techniques. The test is capable of detecting all stiffness-reducing forms of damage such as delamination, composite/honeycomb debonding, as well as assessing the integrity of adhesive joints.

The leaky Lamb wave technique involves immersing a specimen or part under water and passing ultrasonic or Lamb waves through it. As the waves pass through the solid they are deflected and attenuated by defects and damage. Waves, therefore, 'leak' from the sides of the component and are detected by transducers that determine wavespeed and signal attenuation. The technique has been used to measure the degree of matrix cracking in AS4/3502 carbon fibre/epoxy, and has proved to be a useful procedure for evaluating the loss in stiffness of damaged composite parts (74).

The D sight method is used to locate indentations on the surface of damaged composites. The test involves directing light towards the part under examination and obtaining a reflected image on a retroflective screen placed behind the part. Interference of the reflected light waves resulting from surface perturbations such as scratches or dents results in a dark zone on the monitoring screen. The D sight test is capable of detecting delamination in thin laminates as well as surface scratches as small as  $10 \mu\text{m}$  in depth (75). An alternative approach is to paint an impact-sensitive coating on the surface of the composite part. In principle, localized impact loading would create cracks in the surface layer, which could be identified during a general inspection.

Although edge replication is a non-destructive technique, it does not lend itself to the examination of real parts, but remains instead a research tool. The procedure is simple. Impressions are made by softening one side of a cellulose acetate replicating tape with acetone and then placing it on the edge of the specimen. Extreme care is required to ensure that the tape does not slip whilst it is being placed on the surface, otherwise the image will be

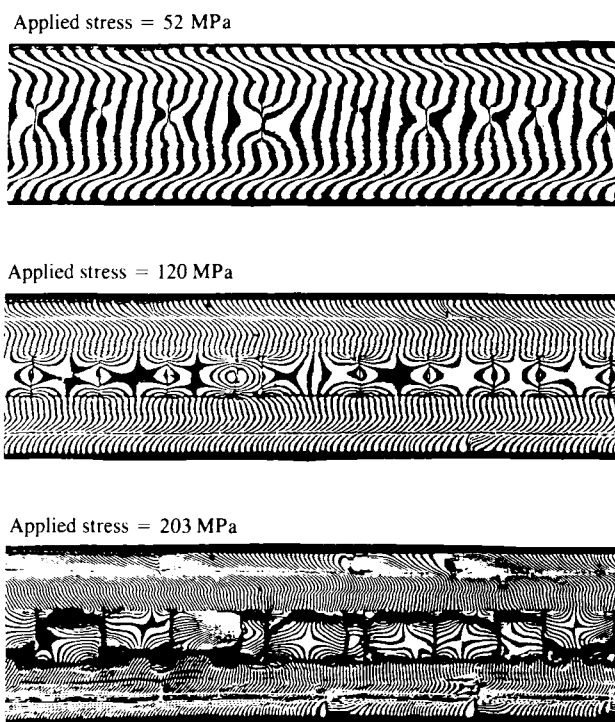


Fig. 15. Displacement contours on the edge of a quasi-isotropic carbon fibre/epoxy composites, at several values of applied load (reference 68)

blurred. The tape is then removed and viewed in transmitted light or in a scanning electron microscope (76). Since cracks tend to close when loading is removed, improved definition of damage can be achieved by leaving the specimens in the test machine and making the replicas under load. Edge replication is a useful procedure for identifying matrix cracking, delamination, and fibre damage along the edges of composite specimens. Clearly, it is not capable of detecting internal damage.

#### 4 DESTRUCTIVE TESTING TECHNIQUES

Although techniques such as thermal deplying and optical microscopy result in either the partial or total destruction of the composite component, they are, nevertheless, useful for determining the cause of failure, as well as establishing the area of crack initiation.

##### 4.1 Thermal deplying

The thermal deply technique has been discussed in detail by Freeman (77). The procedure involves placing the component or specimen in an oven heated to a temperature above that at which the polymeric matrix degrades. For example, for XAS 914C – a brittle carbon fibre/epoxy composite – the optimum deplying temperature is approximately 425°C (31). After a certain time period, typically one hour, the oven is switched off and left to cool. Once at room temperature, the individual plies can be separated using a razor blade and observed for damage. Fibre fracture can be easily identified and quantified using deplied sections. In order to identify zones of delamination, Freeman (77) suggested that a solution of gold chloride solution should be applied to the damage area before deplying. The authors have found, however, that this is not necessary for a brittle matrix carbon fibre composite; the delaminations were clear in untreated deplied sections. The technique is not capable of identifying other damage mechanisms such as matrix cracking but is, nevertheless, useful for examining large areas of fracture such as those introduced during impact loading (31) or more localized regions of failure such as fatigue damage in notched specimens (1).

##### 4.2 Optical microscopy

Optical microscopy is a technique used frequently to identify failure modes in composite materials. Generally, the area to be examined is sectioned, mounted in a potting compound, ground and polished. When viewed under an optical microscope, fibre fracture, matrix cracking, and delamination can be readily identified. The major disadvantages of the technique are that it is laborious and yields only a two-dimensional view of the damage zone. For certain materials it can be difficult to identify matrix cracking in micrographic mounts. One way to overcome this is to add a fluorescent dye to the mounting material (78). When viewed in an optical microscope, the material that has filled up the cracks fluoresces, enabling defects to be identified easily.

##### 4.3 Scanning electron microscopy

A great deal of information can be obtained from the fracture surface of failed components using the scanning electron microscope. For scanning electron microscopy, small specimens are usually coated with a thin layer of gold in order to ensure that the electrons are conducted

away from the area being examined, and, therefore, avoid local heating. Scanning electron microscopy is an excellent post-failure analysis technique and has been used extensively to determine fracture paths in long fibre composites (79). Failure mechanisms such as fibre fracture, delamination, shear yielding, and matrix cracking can be readily identified. Purslow (80) has developed considerable expertise in the area of post-failure analysis of failed composite components. By examining the fracture surface of failed carbon fibres in CFRP, he has been able to identify the direction of crack propagation, and trace the location of crack initiation. Other workers (81) have used fracture surface characteristics in the polymer matrix, such as river marking and hackles, to identify directions of crack propagation, and determine the origins of fracture in fibre reinforced composites.

#### 4.4 Summary

No single test is capable of identifying all of the failure modes observed in composite materials. Ultrasonic C scanning and holography appear to be useful methods for locating and assessing the size of defects. Once this has been achieved, greater information can be obtained using penetrant-enhanced X-ray radiography which is capable of identifying many of the matrix and fibre-dominated modes of failure, and can, if necessary, be adapted to give a three-dimensional appreciation of the damage zone.

#### 5 CONCLUSIONS

Fibre reinforced composites fail in a large number of modes involving the fibres, matrix and interphase region. The relative amounts of each of these, as well as the associated fracture energy, depend upon a large number of parameters, such as the properties of the constituents, the fibre stacking sequence, and environmental conditions. In general, failure modes that involve fracture of the matrix offer low fracture energies, whereas fibre-dominated modes of fracture involve a greater dissipation of energy. The tensile strength of long fibre composites is sensitive to fibre damage, but the compressive properties are influenced by matrix fracture, most particularly delamination.

At present a large number of damage detection techniques are available for evaluating both the nature and severity of damage in composite materials. None of the techniques are capable of detecting all of the various forms of internal damage observed in composite materials. It appears, therefore, that two or more techniques should be used in parallel in order to ensure full detection of damage in these materials.

#### ACKNOWLEDGEMENTS

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#### APPENDIX

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